

Code Class Prompt Sheet

1. Startup a cursor window and then create a new folder and then drag it into the window. Download the Expo Go app. This is a way to test your mobile version of the app on a real mobile device for testing. We also will need to create a Github account so make sure that is set up before we get started.

2. Create a new expo project. Expo is an app framework that lets you build a Web, iOS & Android app all with one codebase. Paste this prompt in your cursor chat.

Prompt: ***Please setup an expo project within this folder (Folder name eg. instagram-app) using Expo SDK 54 set up the project for react typescript please.***

3. Configure the project for the web.

Prompt: ***Please configure the project to run web***

4. Build the Initial User interface based on mockups first attach a mockup. Pinterest is a great resource for whatever application or interface you are looking to build. AI tools like ChatGPT are also great to help design a bespoke interface but it really helps to reference designs made by professional UI/UX designers, Pinterest is a great resource.

Prompt: ***Please design all of the screens for our front end for this application based on the attached design user interface design reference. This is a (Replace this: state the type of app and the goal of the app) style app so it needs all the ui and ux for a professional production ready application. I need all the views for (Replace this: Feed, Explore Page, Direct Messages, Profile Page, Create Post Pop Up Module)***

5. Try the app out with Expo Go make sure the expo go app is downloaded and then open up the camera and scan the QR code that appears after you run the following command

Terminal Command: **npm start**

6. Possible follow ups, after you test the app you will notice things right away so clean these up with some follow up prompts like this.

When you click on a post on explore it should open the post.

When you click on the create post pop up module and select post It should open your device photo library and then you add a photo and then it should open up a real post module to create post, you will add a caption and then create post we should then see this post on the feed and on your profile.

When you click on the comments it should open up a new view that is a view instagram style where you can view the comments and add a new comment when you type a comment the comment should appear.

In the direct messages when you open a conversation it needs to open up a view and when you type a message it should go into this chat

7. Commit to git, it's important to save so you should go to the git changes tab on the left panel. Add a message like initial-front-end-build and then click the ... three dots and then select stage all changes and then hit commit. This is a save point that you can always reference or revert back to.

8. Hook up the backend, now it is time to hook up a cloud computing database using Google Cloud Provider (GCP) firebase so that we can start to save data such as messages, bio information and comments (text based data) as well as posts and profile content (images and video data) we will also setup authentication so that your app can save user information to their profile assign user ID's to all their saved data in the database. Follow these steps to set up your database backend.

Steps:

- a) Go to: <https://firebase.google.com/>
- b) Click on Go to console
- c) Click Create New Firebase Project and give it a name
- d) Click continue
- e) Enable analytics, Create analytics account & Set analytics location
- f) Go to database & Storage on the side panel click firestore
- g) Click create database & select standard edition
- h) Click next on the next screen (nam5)
- i) Start in production mode
- j) Go to rules and change the word false to true (for now test only once live have ai write your permission rules)
- k) On the side panel beside the word spark click upgrade add a card on file (don't worry it won't charge anything until you app has significant scale over 5GB stored and 100GB in downloads per month) Link firebase billing account and set the threshold to \$5
- l) Go to database & Storage on the side panel click Storage Click continue on all pop up screens and then go to rules and change false to true
- m) Go to the side tab and click authentication, click get started click email and password and then enable it and save.
- n) Click the home icon (Project Overview) and click add app select third option (web)
- o) Give your app a name take the
- p) Take the command it gives you and paste it into the terminal npm install firebase
- h) Take the credentials it gives you and paste it into the Cursor chat:

Prompt: ***Please configure the firebase credentials within this project so it's ready to start writing to the backend***

Paste the credentials (example):

```
// Import the functions you need from the SDKs you need

import { initializeApp } from "firebase/app";

import { getAnalytics } from "firebase/analytics";

// TODO: Add SDKs for Firebase products that you want to use
// https://firebase.google.com/docs/web/setup#available-libraries


// Your web app's Firebase configuration
// For Firebase JS SDK v7.20.0 and later, measurementId is optional

const firebaseConfig = {

  apiKey: "AIzaSyDmv_1kpPxuVT4JgfiuUXSI8IXsR1B_-fQ",
  authDomain: "instagram-style-app-78a58.firebaseio.com",
  projectId: "instagram-style-app-78a58",
  storageBucket: "instagram-style-app-78a58.firebaseio.com",
  messagingSenderId: "973074345316",
  appId: "1:973074345316:web:fd5ba48aebb64374f45e71",
  measurementId: "G-HBZ68RJVB9"

};


// Initialize Firebase

const app = initializeApp(firebaseConfig);

const analytics = getAnalytics(app);
```

9. Wire up authentication and have the profile reflect this.

Prompt: ***Please can you wire up email and password login/create account with firebase on app entry and create a proper session store for login persistence. Upon account creation (Create Account) we need to create a collection document in the firestore schema for users that will contain the user ID, Name, Username (@example) Date of birth, bio and profile picture have a professional ui for a login/create account screen. When saving profile picture save the profile picture to storage and save the firebase storage url to the users collection document please***

10. Wire Up App Functionality

a) One shot (if you are feeling extra bold) try and one shot this process the agent should understand the needed backend schema based on the front end design and features we have built (it still helps stating them)

Prompt: ***Please write up all the backend infrastructure for real posts, comments on posts, messages between users (direct messages) Likes on posts followers (All users profile are public no request, accept deny logic) please set up a professional senior developer level schema that makes it the most efficient and scalable please for all of the user data, all posted data, interactions and messages will contain user ID's to be linked to the user. Implement this production ready so all features we have built are fully production ready and have the backend wired up.***

b) Individual features one by one, it may help to try the one shot first because then it can think about the schema as a whole for example a feature like comments and likes are associated with posts so their data should probably be saved as a subcollection branching off of the posts collection. I suggest trying one shot and then test all features, make posts, send messages and you will see on the app if everything is working as expected and you can check the backend and verify all data is being saved. For example I had a bug where The post I made was showing on the feed but not on the user profile

Prompt: ***We need to show a user's posts on their profile posts with their user id should show on this users profile please.***

11.

13. Set Up Expo For Expo.

Prompt: ***Please set up the expo app so that in the package.json we can export the app for ios and android using the commands eas build --platform ios --profile production --clear-cache --message "Build September 2" and eas build --platform android --profile preview --message "Build September 2"***

14. Export Expo Go.

Terminal Command: **eas build --platform ios --profile production --clear-cache --message "Build September 2"**

Terminal Command: **eas build --platform android --profile preview --message "Build September 2"**